Hitesh Bhatnagar

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Education

Vellore Institute of Technology — Vellore, Tamil Nadu
Bachelor of Technology in Electronics and Communication Engineering; ***GPA: 8.37 / 10.0*** — *Sep 2022 – Present*

Experience

Research Intern - Indian Institute of Technology Goa — May 2025 – Jul 2025
Deep Learning & Audio Processing — *Goa, India*

- Pioneered a hybrid **Deep Learning** Active Noise Cancellation (ANC) model by enhancing FxEHCAF with adaptive nonlinear filtering (SFANC), enabling robust suppression under dynamic noise environments.
- Incorporated adaptive learning-rate scheduling, momentum updates, and spectral weight decay in **Python/PyTorch**, significantly improving convergence stability and real-time processing capabilities.
- Designed a lightweight **CNN** for noise-type classification and filter switching, reducing system latency.
- Attained quantifiable performance gains with **MSE = 2.9e-5**, **RMSE = 0.00537**, and **SNR = 27.10 dB**.

IT Risk Advisory Intern - Aumyaa Consulting Services LLP — Feb 2026 – Present
ITGC Audit & AI Solutions Development — *Hybrid / India*

- Executed **ITGC audits** across Access, Change Management, and IT Operations for enterprise clients including **ONGC** and **BMW Financial Services**.
- Performed **user access testing (UAR, SOD, approvals)** ensuring roles were provisioned correctly and post-authorization.
- Analyzed **change requests and job logs**, identifying gaps in approvals, failed job reruns, and monitoring controls.
- Developed and maintained **RCMs and audit testing sheets** with clear evidence mapping and zero-error documentation.
- Built Python-based utilities for PII masking, audit trail review, and anomaly flagging, reducing repetitive manual validation in audit documentation workflows.

Publications

Design of Logistic Distance Metric Based Robust Adaptive Filter for Active Noise Control,
Digital Signal Processing (Elsevier Journal), 2026.

A Novel Three-Layer Hybrid Cryptographic Framework using Enhanced Classical Ciphers,
4th International Conference on Applied Artificial Intelligence and Computing (ICAAIC 2025), IEEE Conference, India, 2025.

Graph-Based Zero-Day IoT Botnet Detection,
Provisionally Accepted, ANRF-Sponsored 2nd International Conference on Next Generation Electronics (NEleX 2026),
Vellore Institute of Technology (VIT), India, 2026.

Projects

High-Performance C++ RAG Engine | *C++, LLM, Llama.cpp, Docker, Linux*

- Developed a **C++ based Retrieval-Augmented Generation (RAG)** system using **Llama.cpp** for efficient local LLM inference on CPU.
- Integrated **4-bit quantized (GGUF) models** and optimized inference using custom sampling techniques (Top-K, Temperature).
- Implemented document retrieval and prompt augmentation pipeline enabling context-aware response generation.
- Containerized the application using **Docker** for reproducible and lightweight deployment.

Embedded RTOS Simulator | *C, Linux, Systems Programming*

- Built a **Real-Time Operating System (RTOS)** simulator in C implementing task scheduling, interrupts, and IPC mechanisms.
- Designed a **priority-based scheduler** with support for task states, context switching, and synchronization using semaphores.
- Developed a CLI-based interface for runtime task control and debugging.

Mini Database Engine in C | *C, Data Structures, DBMS Internals, Systems Programming*

- Built a lightweight **in-memory database engine in C** with a REPL-based SQL-like interface supporting **insert, select, update, and delete** operations.
- Implemented command parsing, fixed-size row storage, execution result handling, and structured control flow inspired by database engine internals.
- Added sorted record insertion and **binary search-based lookup** for efficient ID-based retrieval, update, and deletion.

Technical Skills

Languages: Java, Python, Embedded C, Rust, Bash, Verilog, MATLAB, SQL
Cloud & Network Security: AWS for Cloud Computing, Computer Communication & Networks, Cryptography & Network Security
Frameworks & Libraries: Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, Flutter, Android SDK
Simulation & Hardware: Proteus, Keil uVision, ModelSim, Vivado, NetSim, Cadence, MultiSim, 8051 Microcontroller
Developer Tools: Git, GitHub, Docker, VS Code, Android Studio, IntelliJ, Anaconda, RStudio, Google Colab
Core Concepts: Data Structures & OOP, DBMS, RTOS, DSP, Information Theory and Coding, Probability & Statistics